

Twin Pregnancy 03. Preterm Birth, Cervix, and Prevention

Companion reading handout for simultaneous listening.

Audio course on twin pregnancy. Episode three. Preterm birth, cervix, and prevention limits.

Source base

This episode is based on Gabbe chapters on multiple gestation, preterm labor, and cervical insufficiency, and on the Israeli position paper and local protocols for preterm labor, corticosteroids, magnesium neuroprotection, and PPROM.

Opening frame

Let us begin with the central idea.

Preterm birth in twins is not one disease. It is the final common pathway of uterine stretch, cervical change, membrane vulnerability, fetal and placental disease, and deliberate indicated delivery.

That distinction matters because a prevention strategy that works in one pathway may fail in another.

Twins are among the strongest clinical risk factors for preterm birth. Slightly more than half of twin pregnancies deliver before 37 weeks in many cohorts. But do not stop at the statistic. Ask why.

First, there is uterine overdistension. A uterus carrying two fetuses, more placental mass, and often more fluid reaches stretch states earlier than a singleton uterus. Stretch can alter myometrial gene expression, increase contraction-associated proteins, and influence prostaglandin and inflammatory pathways. The uterus behaves as if term is approaching before the calendar says term.

Second, the cervix may shorten earlier. Transvaginal cervical length is inversely related to preterm birth risk in both singleton and twin pregnancies. A short cervix in twins is therefore a risk marker. But it is not necessarily the same disease as classic cervical insufficiency. It may be a sign that the preterm parturition pathway has already begun.

Third, twin pregnancies have more indicated preterm birth. Preeclampsia, growth restriction, discordant growth, fetal compromise, TTTS, TAPS, sFGR, abruption, and fetal anomaly all push clinicians toward earlier delivery.

Fourth, monochorionicity adds fetal disease. A monochorionic pregnancy may deliver early because one fetus is sick, because the placenta is shared unequally, or because treatment of placental vascular disease creates procedure-related membrane risk.

Now prevention

Now prevention.

This is where residents must resist reflexes.

In singleton pregnancy, progesterone, cerclage, and cervical length algorithms are familiar. In twin pregnancy, the evidence is less generous. Routine prophylactic treatment in twin pregnancy with normal cervical length is not indicated in the Israeli twin follow-up paper. Gabbe describes randomized trials in which progestogens did not reduce preterm birth in unselected multifetal gestations.

The reason is probably mechanistic. If twin preterm birth is driven heavily by uterine overdistension and multiple fetal-placental signals, progesterone alone may not reverse the pathway.

What about a short cervix?

The Israeli guidance allows that vaginal progesterone may have benefit when a twin pregnancy has cervical length less than about 20 to 25 millimeters before 24 weeks. This is not the same as saying every twin gets progesterone. It means the risk phenotype may matter.

Cerclage is even more nuanced.

History-indicated cerclage follows singleton-like indications in the Israeli twin paper. But ultrasound-indicated cerclage for a closed short cervix has no broad consensus. Physical-exam-indicated cerclage may be considered before 24 weeks when there is asymptomatic dilation and no evidence of intrauterine infection.

Gabbe is cautious because some data suggest cerclage can increase preterm birth risk in twin pregnancy with a short cervix. The didactic point is this: in twins, a short cervix may be a marker of global preterm activation rather than isolated mechanical weakness. Placing a stitch into an inflamed or activated system may not fix the cause.

Pessary treatment has not been proven effective in multiple pregnancies in the Israeli twin paper. Activity restriction and bed rest are also not reliable solutions and can increase thromboembolic and deconditioning risks.

Now threatened preterm labor

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Once a twin patient presents with contractions and cervical change, management resembles singleton management in broad categories: assess gestational age, fetal status, infection, membrane status, and maternal stability; consider corticosteroids for fetal lung maturation; consider magnesium sulfate for fetal neuroprotection when delivery before 32 weeks is expected; use tocolysis when appropriate to gain time for steroids or transfer; and use antibiotics when indicated, such as GBS prophylaxis or PPROM pathways.

The local neuroprotection protocol explicitly includes multiple gestation pregnancies. Magnesium is not withheld because there are two fetuses. The indication is expected preterm delivery before 32 weeks.

Corticosteroids are also used similarly in multiple pregnancy when preterm delivery risk is real. But the Israeli twin follow-up paper warns against routine corticosteroids simply because the patient has twins. Steroids are a timed intervention, not a background supplement.

Late-preterm steroids require caution because the evidence base is narrower and often excludes twins in major trials. Local protocols may include twin pregnancy when delivery in the coming week is expected, but this should be used as a clinical decision, not automatic treatment.

Now PPROM, or preterm premature rupture of membranes

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The local PPROM protocol states that rupture of membranes in multiple pregnancy is managed similarly to singleton pregnancy, with cesarean for usual obstetric indications. That is a useful corrective. Twins increase risk, but membrane rupture does not automatically create a separate delivery route.

Now a high-yield medication caveat

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Indomethacin can be useful as a short-course tocolytic before 32 weeks in selected patients, but Gabbe lists twin-to-twin transfusion syndrome as a fetal contraindication. Think physiologically. Indomethacin can reduce fetal urine production and amniotic fluid. In TTTS, amniotic fluid imbalance is already part of the disease, and ductal or renal effects may complicate assessment and safety.

Pause and summarize.

Twin preterm birth is common because the uterus is stretched, the cervix may shorten, membranes are vulnerable, and fetal or placental complications often force

indicated delivery. Prevention is not simply importing singleton algorithms. Routine progesterone is not indicated for all twins. Vaginal progesterone may be considered with a short cervix. Cerclage and pessary require caution. Steroids and magnesium are used when delivery risk and gestational age justify them.

Now final synthesis

Now final synthesis.

Key point

First, twins deliver early for multiple reasons, not one reason.

Second, the cervix is a risk sensor. In twins, a short cervix may reflect preterm pathway activation, not only mechanical failure.

Third, routine preterm-birth prophylaxis in all twins is not supported. Treatment should be risk phenotype driven.

Fourth, magnesium for neuroprotection and corticosteroids remain important, but they are timed interventions for likely preterm delivery, not routine twin medications.

End of episode three.